

Project Design Phase-II

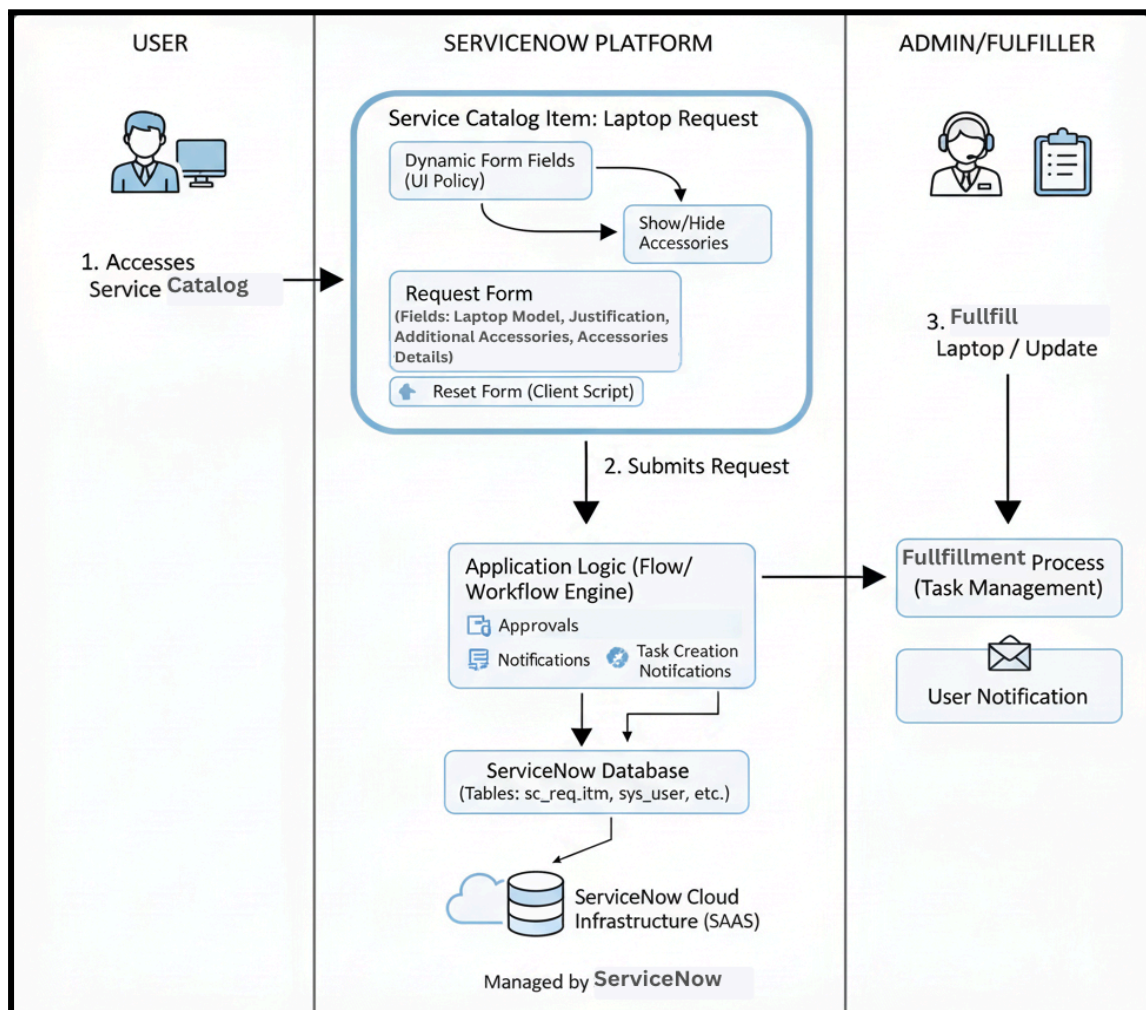
Technology Stack (Architecture & Stack)

Date	01 November 2025
Team ID	NM2025TMID08851
Project Name	Laptop Request Catalog Item
Maximum Marks	4 Marks

Technical Architecture Overview:

The architecture for the **Laptop Request Catalog Item** is built entirely within the ServiceNow platform, leveraging native components for form design, dynamic behavior, governance, and deployment. The solution is designed to be scalable, secure, and user-friendly, with minimal external dependencies.

Technical Architecture Diagram:



Description:

In the context of the ServiceNow platform, the architectural diagram for a Service Catalog item like the Laptop Request is represented by the native **ServiceNow Request Fulfillment process**.

- **User/Employee:** Accesses the **Service Portal** or the standard **ServiceNow Web UI**.
- **User Interface:** Interacts with the **Laptop Request Catalog Item** form.
- **Client Logic (Browser):** **Catalog UI Policy** (for dynamic fields like *Accessories Details*) and **Client UI Action** (*Reset Form*) run in the browser.
- **ServiceNow Cloud/Server:**
 - **Application Logic:** The submission creates a **Request (sc_request)**, **Requested Item (sc_req_item)**, and potentially a **Catalog Task (sc_task)**.
 - **Database:** Request data is stored in the **ServiceNow CMDB/Service Catalog tables** (e.g., *sc_item_option*, *sc_req_item*).
 - **Notifications:** (If configured) Notifications are sent via **ServiceNow Notifications** upon submission or fulfillment.
- **External Interface:** (Optional) Integration with an **HRMS/User Directory** to verify employee details.

Table-1: Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	End-users interact with the form via the Service Portal or the main application. Admin/Developer interacts via the standard UI.	ServiceNow Service Portal / ServiceNow Web UI
2.	Application Logic-1	Defines the structure, variables, and categories of the catalog item.	ServiceNow Service Catalog / Variables
3.	Application Logic-2 (Client-Side)	Implements dynamic form behavior based on user input (e.g., showing/hiding fields).	ServiceNow Catalog UI Policy
4.	Application Logic-3 (Client-Side)	Provides user-focused action like clearing the form fields.	ServiceNow Client UI Action (Script)
5.	Database	Stores the submitted request data, including variable values.	ServiceNow CMDB / Service Catalog Tables
6.	Cloud Database	Managed backend database for the entire ServiceNow instance.	ServiceNow Cloud Database

7.	File Storage / Tracking	Captures the configuration changes for governance and deployment.	ServiceNow Local Update Set / XML Export
8.	External API-1	(Optional) HRMS integration to verify the user's role or eligibility.	REST API in ServiceNow
9.	Infrastructure (Server / Cloud)	The entire solution is hosted and managed on the platform.	ServiceNow Cloud (SaaS) Platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Not applicable (ServiceNow is proprietary).	-
2.	Security Implementations	Access control for the catalog item, variables, and update set tracking are role-based.	ACLs, Scoped Applications, Update Set History
3.	Scalable Architecture	The solution is built on a SaaS platform designed for high horizontal scalability.	ServiceNow Cloud Architecture (SaaS)
4.	Availability	Highly available via the ServiceNow cloud hosting and architecture.	Load-balanced ServiceNow Instances
5.	Performance	Optimized by using out-of-the-box ServiceNow features (UI Policy, Client Action) rather than extensive custom scripts.	Native ServiceNow features (UI Policies), Client Scripting